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DATA SET CHANGE NOTICE

Reference No:	DSCN 02/2010
Version No:	1.0
Subject:	Date and Time Input for Clinical Systems within the NHS in England
Type of Change:	Introduction of a new approved Information Standard
Implementation Date:	01 December 2015
Business Justification:	To make display, input and user interaction consistent across all clinical systems within the NHS in England in order to increase patient safety by maximising clinical utility and minimising reading and transcribing errors.

Introduction

Displaying and inputting unambiguous information in a consistent format is a core element in ensuring effective patient care. It is vital that healthcare IT users correctly interpret and input information relating to all aspects of patient care and management.

This user interface standard defines the valid input rules for fully specified dates and time, partially specified dates, approximate dates and times, arithmetic shortcuts, date and time durations, free text fields, controls and disambiguation of date entries.

The standard has been developed as part of the Common User Interface (CUI) programme within the Technology Office of the Department of Health Informatics Directorate. This programme is addressing the safety risks posed by variation across the NHS in the way information is input and displayed. Its aim is to support healthcare IT users in correctly inputting and interpreting information relating to all aspects of patient care and management.

Background

The input of date and time information to clinical and healthcare IT systems is a fundamental function within clinical and administrative processes. It is vital that healthcare workers correctly enter dates relating to patient demographics, clinical episodes and planned treatments, among others. Promoting patient safety is of primary importance and depends upon the removal of ambiguity during date and time input. Therefore, it is vital that date and time input controls provide a clear mechanism for guiding users towards the entry of an accurate value, while also assisting the user to disambiguate entries when necessary.

Examples of use include:

- Specific drug administration dates within a course of medication.
- Identification of a patient using a date of birth.
- Key appointment dates and times within a patient care plan.

In almost all cases, errors in the entry of date and time information will have safety implications for patients.

Modern user interfaces offer the user a variety of mechanisms for entering dates and times. These range from constrained methods, such as drop-down lists providing a small set of allowable values, to more flexible calendar and spin controls, and highly flexible free-text entry fields. Each method exhibits strengths and weaknesses, and can be appropriate for specific purposes.

The objective of this standard is to provide a consistent, safety-assured method of entering dates and times. A precondition for this is compliance with the Common User Interface standard for Date Display, as the short format described in this document is used by the date input control. Likewise, a further precondition is compliance with the Common User Interface standard for Time Display, as this format is used by the time input control. These standards must be used in all cases where the NHS application displays dates or times that affect patient treatment and identification. The entry of dates in the long date format is currently under review.

Details of the Standard

The standard is published with associated guidance in a Common User Interface Design Guide. It is applicable to user interfaces displayed on desktop or laptop computers. It is assumed that, at a minimum, these computers are capable of operating at a minimum display resolution of 1024 x 768, and have a keyboard and pointing device.

The following items are covered in the guidance:

- Fully specified dates.
- Fully specified times.
- Partially specified dates.
- Approximate dates and times.
- Arithmetic shortcuts.
- Date and time durations.
- Input via free text fields.
- Input via input controls.
- Disambiguation of date entries.

Aspects that are **out of scope**:

- Date and time display.
- Default dates.
- Durations shorter than a second.
- Synchronisation.
- Data storage
- Display styles.

Safety Case

Clinical safety requirements and objectives related to the input of dates and times in clinical systems are documented in the safety case that accompanies this standard (www.isb.nhs.uk/documents/cui). Organisations that develop, supply, deploy and use clinical systems are required to identify and manage any risks in current systems until the standard can be implemented. They must also mitigate those live implementation risks outlined in the safety case for which responsibility for mitigation has been transferred to the supplying or deploying organisation.

The process for assessing and managing safety risks is detailed in NHS information standards for the application of patient safety risk management to the manufacture, deployment and use of health software (<http://www.connectingforhealth.nhs.uk/dscn/dscn2009/data-set-change/>), i.e.:

- Application of Patient Safety Risk Management to the Manufacture of Health Software (DSCN 14/2009); and
- Application of Patient Safety Risk Management to the Deployment and Use of Health Software (DSCN 18/2009).

Conformance

For full conformance to be demonstrated, the elements marked as 'Mandatory' in the Common User Interface Design Guide must be complied with and there must have been mitigation of those live implementation risks outlined in the safety case for which responsibility for mitigation has been transferred to the supplying or deploying organisation.

Timescales for Implementation

FRAMEWORK	Health and Social Care Personnel	Organisation	IT Suppliers
Effective Date “may use”	Immediate	Immediate	Immediate
Implementation Date “must use”	When available in system	At next major system release	At next major system release
Conformance Date “must be used effectively and assessed for use”	01 December 2015	01 December 2015	01 December 2015
Superseded Date (of prior standard) “stop using prior standard”	Not applicable	Not applicable	Not applicable

Effects on Other Information Standards:

None

Sponsor Details

Ken Lunn, Director of Data Standards and Products
Department of Health Informatics Directorate
Technology Office

Further Information and Support

- CUI Date and Time Input Design Guide - www.isb.nhs.uk/documents/cui
- Clinical Safety Case - www.isb.nhs.uk/documents/cui
- CUI website - www.cui.nhs.uk
- Standards documentation – www.isb.nhs.uk/documents/cui
- Queries - cui stakeholder@mailbox@nhs.net

Appendix

This DSCN must be read in conjunction with the Clinical Safety Case and Closure Report for Demographics / Information Input and Display, which can be found at the following link: www.isb.nhs.uk/documents/cui

Table Notes:

1. Relevant organisations are those organisations as defined in the standard who must take direct action to implement the standard
2. IT Suppliers are all suppliers to the organisations listed at ¹ who supply functionality pertinent to that standard
3. **Effective Date** is the date from which a new standard can be used but may not be mandatory. This might facilitate piloting, for example, or enable time for system functionality development. At this point, **you “may use” the standard.**
4. **Implementation Date** is the point from which the new standard becomes mandatory. Ideally, it inherently implies organisations use appropriate systems i.e. the date is the same for organisations and suppliers. However, there may be circumstances where interim workarounds are required i.e. the date is different for organisations and suppliers. At this date, **you “must use” the standard.** Where the standard demands data is submitted centrally, sub components of implementation date (and possibly ‘effective date’) are:
 5. **Collection Start Date** – this is the date collection of data must begin
 6. **First Submission Date** – this is the date of first submission of data centrally
 7. **Reporting Period / Submission Cycle** – If the standard calls for further collection and submission at defined intervals, this cell provides text of the reporting period (e.g. calendar month, financial year)

and the submission cycle (e.g. submit data monthly on the 10th working day of the subsequent month).

8. **Conformance Date** is the date from which the service and IT system suppliers must use the standard as envisaged i.e. using appropriate IT solutions rather than interim workarounds and, if the standard requires it, an independent, authoritative body or legitimate internal audit would conduct a conformity assessment with the expectation of full conformance by all relevant parties. It is the ***“must use standard effectively and assessed for use”*** date
9. **Superseded Date** of the prior standard sets the date at which the prior standard is replaced by the new standard i.e. the prior standard must no longer be used. This date will apply only where there was a pre-existing standard made redundant by the new standard. It might be different from preceding dates in the framework if, for example, a new and old standard run in parallel for a period. It is the date from which you ***“stop using the prior standard”***.